

MIT INVITATIONAL 2026 — ROCKS & MINERALS

WORKED ANSWER KEY WITH EXPLANATIONS

Reading note: Questions marked [SPEC] depend on physical specimens. For those I give the identification that the station's own internal logic forces (theme, follow-up questions, diagrams), plus the reasoning chain. Everything else is answered outright.

STATION 1 — "OF QUARTZ YOU LOVE ROCKS!" [23 PTS]

1. Identify specimens A-D [SPEC, 4 pts]

The station is a sandstone/quartz-maturity station. The follow-ups pin the specimens down:

SPEC	ALMOST CERTAINLY	WHY THE EXAM FORCES IT
A	Quartz arenite (quartz sandstone)	Q2a says it plots at the Q apex; Q3 asks "why is A's composition so <i>pure</i> "
B	Arkose	Q2b asks which <i>other</i> letter shares its compositional definition → a feldspar-defined rock
C	Granite (or a felsic pluton rock)	Q6 asks where C formed on a <i>pluton</i> cross-section
D	Conglomerate / breccia or shale	the remaining clastic slot

2. QFL (Q-F-L) triangle

(a) Where does A fall? → A. Top Quartz arenite is ≥ 90 –95 % quartz, so it plots against the Q apex.

(b) Which letter could be specimen B? → C (E is also defensible) Arkose is *compositionally* defined as a sandstone with **>25 % feldspar and F > L**. Point C sits at roughly 40 Q / 60 F — squarely arkosic. Point E (≈ 90 F) is also arkose by composition; if the exam wants a single answer, C is the safer pick because E is so feldspar-dominant it edges toward a feldspar-rich grus. The point being tested: the *name* comes from the composition, not from the specimen's appearance.

(c) Highest ZTR index corner? → A. Top ZTR (zircon-tourmaline-rutile) is a **maturity** index. Ultra-stable heavy minerals concentrate in the same sediments that have lost their feldspars and lithics — i.e., the quartz-rich corner. High Q = high maturity = high ZTR.

(d) Depositional environment for high-ZTR rocks [3 pts] A **high-energy, long-transport, repeatedly reworked** setting — beach/shoreface, or an aeolian (desert dune) system, typically on a tectonically quiet passive margin or stable craton. *Why:* Zircon, tourmaline, and rutile are the "ultrastable" heavy minerals: hard (7–7.5), chemically inert, and cleavage-poor. In a setting with prolonged abrasion, winnowing, and chemical weathering, feldspars hydrolyze to clay, micas and amphiboles break down, and lithics disintegrate — but ZTR grains survive. Every additional cycle of erosion and redeposition therefore *enriches* the residue in ZTR. It's an index of survivorship, not of source abundance.

(e) Origin of the sediments [2 pts] A **felsic intrusive igneous rock — granite / granitic pegmatite** (a quartz + K-feldspar-rich, peraluminous plutonic rock). Zircon is the classic accessory of Zr-saturated felsic melts; tourmaline requires boron, which concentrates in late-stage granitic fluids and pegmatites. Mafic rocks give you olivine and pyroxene, not zircon and tourmaline.

3. Why is A's composition so pure? [2 pts]

Two mechanisms, and full credit almost certainly wants both: - **Chemical:** quartz is essentially inert to hydrolysis at surface conditions; feldspars and ferromagnesian minerals weather to clays and oxides and are flushed away. - **Physical:** quartz has H = 7 and no cleavage, so it resists abrasion; softer/cleaved grains are ground to silt and mud and winnowed out. Add **recycling** — a supermature quartz arenite is usually second- or third-cycle sand, already stripped of everything unstable before its final burial.

4. Sedimentary structure in E [SPEC, 2 pts]

Almost certainly **ripple marks**. The diagnostic: - **Asymmetrical** (steep lee side, gentle stoss side) = **current** ripples, unidirectional flow (river, tidal channel). - **Symmetrical** = **oscillation/wave** ripples, bidirectional flow (shoreface). If the specimen is a hand sample with a gentle slope on one side and a steep one on the other, answer *asymmetrical, current-formed*.

5. Dune type and flow direction [2 pts]

The aerial photo shows crescent-shaped dunes with trailing tips → **barchan dunes** (crescentic dunes). Barchans form in unidirectional wind with limited sand supply. **The horns point downwind**, the gentle slope faces upwind, and the steep slip face is on the downwind (lee) side — so read flow direction as *from the convex/gentle side toward the horns*.

6. Where on the pluton diagram did C form? [2 pts]

Specimen C is the granite. It crystallized in the **magma body itself (region C, the red chamber)** — the **magma chamber / pluton**, which at this scale would be called a **batholith or stock**. Worth knowing for partial credit: the dashed ring (D) surrounding the chamber is the **contact aureole**, the zone of contact metamorphism; F is a **dike**; A is the overlying country rock/roof.

7. α vs β quartz at 600 °C [2 pts]

The $\alpha \leftrightarrow \beta$ inversion is at **573 °C at 1 atm** (it rises with pressure, but at shallow crustal levels 573 °C is the working number). - At 600 °C in the crust it was **β -quartz** (high quartz, hexagonal, point group 622). - Sitting in the exam room it is **α -quartz** (low quartz, trigonal, 32). The inversion is **displacive** — bonds bend rather than break — so it is instantaneous, unquenchable, and *reversible*. A crystal that grew as β retains β 's hexagonal external morphology (bipyramidal "beta quartz" habit) even though its internal structure is now α .

STATION 2 — "UNDER PRESSURE" [21 PTS]

This is a **coal rank series** station keyed to Pennsylvania.

1. Identify A-C and order by formation [SPEC, 4 pts]

The rank series is **peat → lignite → sub-bituminous → bituminous → anthracite**. Q3 ("what color does B burn as") and Q4 ("lowest moisture") both point at **B = anthracite**. A likely set: A = lignite, B = anthracite, C = bituminous → **formation order A → C → B** (lignite, bituminous, anthracite).

2. Geologic period [1 pt]

The **Carboniferous** — in North American usage, the **Pennsylvanian** subperiod (~323–299 Ma). Extensive equatorial coal swamps, high atmospheric O₂, and lignin-producing trees that fungi could not yet efficiently decompose.

3. Color B burns [1 pt]

Blue. Anthracite is >90 % fixed carbon with almost no volatiles, so it burns with a short, nearly smokeless **blue flame** rather than the yellow, sooty flame of volatile-rich bituminous coal.

4. Lowest moisture by weight [1 pt]

B, anthracite (~2–5 % moisture, vs. lignite at 30–45 %). Moisture drops monotonically with rank as burial expels pore water.

5. Pennsylvania sub-questions

(a) → C. Mica schist. Two Appalachian orogenies in that window (Taconic ~450 Ma, Acadian ~380 Ma, Alleghanian ~325–260 Ma) subjected a thick **pelitic** (shale/mudstone) passive-margin wedge to regional Barrovian metamorphism. Pelites → slate → phyllite → **schist** → gneiss. Quartzite needs a quartz-sand protolith, marble a carbonate one, amphibolite a mafic one; none of those is the dominant protolith of the PA Piedmont.

(b) Mineral responsible for the texture [1 pt] Mica (muscovite and biotite). Their sheet (phyllosilicate) structure grows with (001) perpendicular to maximum compressive stress, producing the strong planar preferred orientation = **schistosity**, and the characteristic silvery sheen.

(c) Identify D and E [SPEC, 2 pts] Almost certainly **shale and sandstone** — the other two members of a Pennsylvanian **cyclothem**, and the two rocks Q7/Q8 go on to discuss alongside limestone.

(d) Which grains are older? [1 pt] The sandstone's grains. This is a nice conceptual trap. The quartz (and ZTR) grains in sandstone are *inherited* — detrital survivors eroded from far older source rocks, potentially recycled through several previous sandstones. The clay minerals that make up shale are largely **neofomed/authigenic**: they are *created* by hydrolysis of feldspar during the weathering event itself. So the sand grains predate the shale's clays.

(e) Arrangement in the basin [2 pts] The setting is a **foreland basin** compressed on both sides and later folded into a **synclinerium** (the Anthracite Region). Predict: **cyclic interbedding** — repeating packages of sandstone → shale → coal → (limestone) — with the coal seams sandwiched between thick clastic layers and the whole stack **folded into tight synclines**.

Why that explains PA's anthracite: (i) the clastic overburden supplied deep burial → high temperature and pressure; (ii) Alleghanian compression added tectonic strain and heat, driving coalification past bituminous to **anthracite**; (iii) synclinal keels *preserved* the seams from erosion while anticlinal crests were stripped.

(f) Order-of-magnitude of PA anthracite reserves [1 pt] Remaining recoverable anthracite is on the order of a few billion short tons ($\sim 5\text{--}7 \times 10^9$ tons). $5 \times 10^9 \text{ tons} \times 2000 \text{ lb/ton} \approx 10^{13} \text{ lb} \rightarrow \sim \mathbf{10^{13} \text{ pounds}}$ (10 trillion). Anything from $10^{12}\text{--}10^{13}$ should score.

6. East or West PA has more schist? [1 pt]

East. The map shows anthracite in the east and bituminous in the west. Higher coal rank = higher T/P = closer to the orogenic core. The Appalachian metamorphic front (the Piedmont/Reading Prong) lies to the east, so schist tracks with anthracite.

7. Weathering susceptibility [2 pts]

Most → least susceptible: **shale > limestone > sandstone**. - Shale: fissile, high porosity along partings, clays swell/shrink, mechanically disaggregates fast. - Limestone: chemically very vulnerable (carbonic acid dissolution) but mechanically coherent — dominant in humid climates. - Sandstone: quartz grains are inert and often silica-cemented → most resistant.

8. Ridges vs. valleys [3 pts]

Ridges: sandstone (and any conglomerate) — resistant, quartz-cemented, low solubility, holds up topography (e.g., the Tuscarora and Bald Eagle sandstones of the Valley-and-Ridge). **Valleys: shale and limestone** — shale erodes mechanically, limestone dissolves; both are stripped down, producing the linear valleys (and, in the Great Valley limestones, karst). This is **differential erosion** on plunging folds: the map-view zigzag ridges are the resistant sandstone units tracing out fold hinges.

STATION 3 — QAPF [15 PTS]

1. Intrusive or extrusive easier to classify? [1 pt]

Intrusive. QAPF classification is a **modal** (volume-%) scheme, and it requires you to actually measure how much Q, A, P, and F you have. Intrusive rocks are **phaneritic** — slow cooling makes crystals large enough to identify and point-count by eye or thin section. Extrusive rocks are aphanitic or glassy; their modes are often unmeasurable, which is why volcanic rocks are usually classified chemically instead (TAS diagram).

2. What the vertices stand for [2 pts]

Q = Quartz (and silica polymorphs), **A = Alkali feldspar** (orthoclase, microcline, sanidine, anorthoclase, albite $\text{An}_{0\text{--}5}$), **P = Plagioclase** ($\text{An}_{5\text{--}100}$), **F = Feldspathoid ("foid")** — nepheline, leucite, sodalite, analcime. Note the diamond's logic: Q and F are **mutually exclusive** (silica-oversaturated vs. undersaturated); a rock cannot contain both stable quartz and stable feldspathoid, because they react (e.g. $\text{NaAlSi}_3\text{O}_8 + 2\text{SiO}_2 \rightarrow \text{NaAlSi}_3\text{O}_8$).

3. Which minerals belong to Q, A, P, or F? [3 pts] → A, D, E

- **A. Citrine — YES (Q).** A variety of quartz colored by Fe^{3+} .
- **B. Tourmaline — no.** Cyclosilicate, accessory only.
- **C. Augite — no.** Clinopyroxene; a mafic mineral, counted in "M," not in QAPF.
- **D. Labradorite — YES (P).** Plagioclase, $\text{An}_{50\text{--}70}$.
- **E. Sodalite — YES (F).** Classic feldspathoid.
- **F. Beryl — no.** Cyclosilicate.

4. Largest surface area intrusion [1 pt] → D. Batholith

By definition $>100 \text{ km}^2$ of exposed area; laccoliths, lopoliths, dikes and stocks are all smaller (a stock is $<100 \text{ km}^2$, which is precisely the dividing line).

5. Texture of specimen A [SPEC, 1 pt]

If A is a granite (see Q6), the expected answer is **phaneritic** (coarse-grained, equigranular). If it has two distinct crystal-size populations, **porphyritic**; if giant crystals, **pegmatitic**.

6-10. Placing specimens [SPEC]

Here is the numbered-field key you need — with it, any specimen you can name maps to a field:

FIELD	ROCK	FIELD	ROCK
1 / 1a	quartzolite / quartz-rich granitoid	9	monzodiorite / monzogabbro
2	alkali feldspar granite	10	diorite / gabbro / anorthosite
3 (3a/3b)	granite (syenogranite/monzogranite)	11	foid syenite
4	granodiorite	12	foid monzosyenite
5	tonalite	13	foid monzodiorite
6	alkali feldspar syenite	14	foid gabbro / foid diorite
7	syenite	15	foidolite (e.g. ijolite)
8	monzonite		

Most likely intent: **A (granite) → 3; B (gabbro or diorite) → 10; C (granodiorite) → 4; D → 2 or 6** depending on whether it's quartz-bearing; **E (nepheline syenite or foid-bearing) → 14 or 6** if it's an alkali feldspar syenite. Boundary rule to memorize: the vertical divisions in the **upper** half are at $A/(A+P) = 10, 35, 65, 90$; in the **lower** half only at 10, 50, 90.

11. 65 % Q, 25 % A → B. 3

$Q = 65 \rightarrow$ in the 60–90 band (fields 2–5). Remaining 35 % feldspar splits 25 A / 10 P, so $A/(A+P) = 25/35 = 71 \%$, which falls in the 65–90 band = **granite (field 3, specifically 3a syenogranite)**.

12. 15 % P, 33 % F → D. 12

$F = 33 \rightarrow$ lower half, foid-bearing. With $Q = 0$, $A = 100 - 33 - 15 = 52$. $P/(A+P) = 15/67 = 22 \%$, so $A/(A+P) = 78 \%$, which lands between the 50 and 90 divisions → **foid monzosyenite (field 12)**.

STATION 4 — UV REACTIVE [24 PTS]

1. Identify the fluorescent minerals A-F [SPEC, 7 pts]

Internal clues nail two of them: D is a zinc **silicate** from a famous US zinc state → **willemite (Zn_2SiO_4)**; E is a **silicate** that is blue in daylight → **sodalite/hackmanite**. The remaining slots are the standard fluorescent suite: **calcite** (red-orange, Mn^{2+}), **fluorite** (blue-violet, Eu^{2+}/REE), **scheelite** (blue-white, intrinsic WO_4), **aragonite**, **hydrozincite** (blue-white), **autunite** (yellow-green, uranyl).

2. Specimen E under UV [3 pts]

- Color: **orange** (bright orange-red).
- Element causing the fluorescence: **sulfur**.
- Ion causing the blue color in normal light: the **trisulfur radical anion, S_3^-** (a color center trapped in the sodalite cage; the same species colors lapis lazuli). Bonus context: hackmanite (S-rich sodalite) also shows **tenebrescence** — it darkens under UV and fades in daylight.

3. Adamite [3 pts]

- Color under UV: **green** (bright yellow-green).
- Causative ion: the **uranyl ion, UO_2^{2+}** . Uranyl gives a distinctive structured green emission (a vibronic progression from the symmetric $O=U=O$ stretch) and is exactly what makes **hyalite opal** glow green, and what makes uranium glass glow. Note that uranyl is a trace *impurity* here, not a formula component of Zn_2AsO_4OH .

4. Silicate classes of D and E [2 pts]

- **D (willemite): nesosilicate** — isolated SiO_4 tetrahedra, $Si:O = 1:4$, linked by Zn.
- **E (sodalite): tectosilicate** (framework), specifically a **feldspathoid** — a fully polymerized aluminosilicate cage framework with Cl^-/S enclosed.

5. Hardest mineral A-G, inferred not scratched [1 pt]

Adamite (Zn_2AsO_4OH) is $H \approx 3.5$. The reasoning the question wants: hardness scales with **bond strength and structural connectivity**. Ranking the likely suite: calcite 3, adamite 3.5, fluorite 4, scheelite 4.5–5, willemite 5.5, **sodalite 5.5–6**. So the

answer is **the sodalite (E)**, with willemite (D) as the close runner-up — the two framework/network silicates, because Si-O bonds are far stronger than the ionic Ca-CO₃ or Zn-AsO₄ linkages.

6. State famous for zinc mines of willemite [1 pt]

New Jersey — Franklin and Sterling Hill, Sussex County, the most famous fluorescent mineral locality on Earth (willemite green + calcite red).

7. Specimens with three-fold symmetry [2 pts]

Calcite and willemite — both are **trigonal** (calcite R3c, willemite R3), so both have a 3-fold axis along c. Careful with fluorite: it's isometric (Fm3m) and does technically possess four 3-fold axes along the body diagonals, so it can be argued in; but the intended answer is the two trigonal minerals.

8. Jablonski diagram

(a) → A. A = Fluorescence, B = Phosphorescence. Line A drops from the **singlet excited state S₁** straight to S₀ — spin-allowed, nanosecond lifetime = **fluorescence**. Line B originates from **T₁**, reached via **intersystem crossing** (the pink squiggle), and the T₁ → S₀ transition is spin-forbidden, so it is slow (ms to hours) = **phosphorescence** — that's the afterglow you see when the lamp goes off. Consistent with the colors: B starts lower on the energy axis, so it's the longer-wavelength, redder emission.

(b) No. [2 pts] Fluorescence and phosphorescence in minerals are **activator- and defect-dependent**, not fixed properties of a mineral species. Phosphorescence requires that excited electrons get trapped — either in a triplet state or, more commonly in minerals, in **lattice defect traps** — and then be released slowly. Whether those traps exist depends on which impurity ions are present, in what concentration, and what point defects the crystal has, all of which vary locality to locality. Two calcites can both fluoresce red from Mn²⁺ while only one phosphoresces. **Concentration quenching** and competing "killer" ions (notably Fe²⁺) can suppress one and not the other.

(c) Not always. [2 pts] *Within a single material*, yes: T₁ lies below S₁, so that substance's phosphorescence is red-shifted relative to its fluorescence. But the claim fails as a general rule **across different materials** — a wide-band-gap phosphor's phosphorescence can easily be at a shorter wavelength than a narrow-gap material's fluorescence. Even within one specimen, both emissions are **broad bands** with vibrational structure, so the tails overlap; and thermally assisted delayed fluorescence can return population from T₁ to S₁ and emit *at the fluorescence wavelength* on a phosphorescence timescale.

STATION 5 — "THINGS THAT ARE... PRECIOUS" [26 PTS]

1. Identify A-J [SPEC, 12 pts]

An ore/economic-mineral station. Expect from: **gold, silver, copper (natives); pyrite, chalcopyrite, bornite, galena, sphalerite, cinnabar, molybdenite (sulfides); magnetite, hematite, chromite (oxides); bauxite; quartz** as gangue.

2. Chiral mineral [1 pt]

Quartz — α-quartz crystallizes in enantiomorphic space groups P3₁21 and P3₂21 (right- and left-handed screws of SiO₄ tetrahedra), which is why it is optically active and shows trigonal trapezohedral faces. **Cinnabar** is the other classic answer (P3₁21/P3₂21, and the most optically active mineral known) — if there's cinnabar at the station, it's an equally correct response.

3. Magnetite and magnetism

(a) Magnetite, Fe₃O₄. (b) [3 pts] Fe(A) = tetrahedral sites, occupied by Fe³⁺. Fe(B) = octahedral sites, occupied by an equal mix of Fe²⁺ and Fe³⁺. Formula written structurally: Fe³⁺[Fe²⁺Fe³⁺]O₄. (c) [2 pts] A : B = 1 : 2, and the structure is the **inverse spinel. (Normal spinel would put the divalent cation in the tetrahedral site; magnetite "inverts" it because Fe³⁺ with d⁵ has zero crystal-field stabilization energy in either site, while Fe²⁺ d⁶ gains CFSE in the octahedral site.) (d) [1 pt] **The B sites**. There are twice as many of them, so the B sublattice moment exceeds the A sublattice moment and the net magnetization points with B. In fact the two Fe³⁺ moments (one on A, one on B) cancel exactly, and the entire net moment of magnetite — ~4 μB per formula unit — is the **leftover octahedral Fe²⁺**.**

4. Most energy to reduce by electrolysis [1 pt]

Whichever specimen carries the **most electropositive metal** — the one with the most negative standard reduction potential and largest Gibbs free energy of formation. If **bauxite/aluminum** is present, that's the answer (the Hall-Héroult process is

famously ~13–15 kWh/kg, which is why aluminum was once more precious than gold). Otherwise **sphalerite (Zn)**. Gold and copper are trivially easy by comparison — natives need no reduction at all.

5. Habit and luster of H [SPEC, 2 pts]

Fill from the specimen. Common answer pairs at this station: native copper/silver = **dendritic/arborescent habit, metallic luster**; pyrite = **cubic/striated habit, metallic**; hematite = **botryoidal/reniform, metallic to submetallic (earthy if ochre)**; galena = **cubic, metallic**.

6. Environment specimen F formed in [1 pt]

Most likely **hydrothermal vein** (epithermal or mesothermal) if it's a sulfide or native metal; **magmatic/orthomagmatic cumulate** if chromite or magnetite; **supergene/weathering zone** if it's an oxidized secondary mineral (malachite, bauxite).

7. Most weight-efficient to mine [2 pts]

The **native metals** — native gold, silver, or copper, at essentially **100 % metal by mass**. Nothing with an anion can compete. Among the compounds, run the numbers: galena PbS = 86.6 % Pb; magnetite = 72.4 % Fe; hematite = 69.9 % Fe; chalcopyrite = only 34.6 % Cu. So the ordering to quote is: natives >> galena > magnetite > hematite. (Worth a sentence of nuance for full credit: *weight-efficiency of the mineral* is not the same as *economic grade* — that depends on ore concentration in the rock and on processing cost.)

STATION 6 — POLYATOMIC ANIONS [22 PTS]

1. Identify A-G [SPEC, 7 pts]

Carbonates + sulfates + phosphates. The internal clues: B and D are the **dolomite-ankerite** pair; C and E are the **barite-celestine** pair (Q5 says anglesite is in their group); two specimens are **calcite and aragonite** (Q8a). Add **gypsum/selenite** (Q7) and likely **malachite/azurite or apatite** for G.

2. Formula of the Fe-bearing mineral, and why Fe replaces that ion [3 pts]

Ankerite, $\text{Ca(Fe,Mg)(CO}_3)_2$, ideally **$\text{CaFe(CO}_3)_2$** — dolomite $\text{CaMg(CO}_3)_2$ with Fe^{2+} substituting for **Mg²⁺**, not Ca^{2+} . Justification via **Goldschmidt's rules**: substitution requires matching **charge** and **ionic radius within ~15 %**. - Charge: Fe^{2+} , Mg^{2+} , Ca^{2+} are all 2+, so charge doesn't discriminate. - Radius (VI-coordinate): $\text{Mg}^{2+} = 0.72 \text{ \AA}$, **Fe^{2+} (high spin) = 0.78 \AA** — an 8 % mismatch, easily tolerated. $\text{Ca}^{2+} = 1.00 \text{ \AA}$ — a 22 % mismatch with Fe^{2+} , outside the limit. Structurally, dolomite has **two distinct octahedral sites** of different size alternating along c; Fe^{2+} goes into the smaller Mg site. The dolomite-ankerite series is continuous only up to ~70 % of the Fe endmember.

3. Most vigorous acid reaction [1 pt]

The **calcite** (or aragonite if present — see below). Pure CaCO_3 effervesces vigorously in cold dilute HCl. Dolomite reacts only weakly unless powdered, because the ordered alternating Ca/Mg layers and stronger Mg–O bonds slow the dissolution kinetics. Aragonite, being the *metastable* and slightly more soluble polymorph, reacts marginally faster than calcite — so "aragonite" is the technically sharpest answer.

4. Blue color in C [2 pts]

C is the sulfate in the barite group that's characteristically pale blue: **celestine (SrSO_4)**. The commonly cited cause is trace **gold** — colloidal Au nanoparticles dispersed in the crystal (the same mechanism that makes ruby glass red-to-blue depending on particle size); some sources instead attribute it to **sulfur color centers** produced by natural irradiation. Either explanation should be accepted with reasoning. **Without the impurity, celestine is colorless to white**. (If your specimen C is instead blue **barite**, the same logic applies — the $\text{Ba}^{2+}/\text{Sr}^{2+}/\text{SO}_4^{2-}$ ions have no partly filled d-shells, so the mineral has no intrinsic color and any hue must be an impurity or defect.)

5. Anglesite composition [1 pt]

PbSO₄ — lead(II) sulfate, isostructural with barite, celestine, and anhydrite. It's the classic supergene oxidation product of galena ($\text{PbS} + 2\text{O}_2 \rightarrow \text{PbSO}_4$).

6. Pair forming a solid solution at room T [1 pt]

Barite (BaSO_4) - celestine (SrSO_4) — specimens C and E. Ba^{2+} (1.42 Å, XII) and Sr^{2+} (1.44 Å, XII) are nearly identical in size and charge and the two are isostructural, so they form a **complete solid solution** at ambient conditions. Contrast: calcite–

magnesite needs high T (Ca/Mg radius mismatch is too large at room temperature, which is exactly why ordered dolomite exists), and dolomite-ankerite is only partial.

7. Dana class where selenium is found [1 pt]

Class 02 — the sulfides (sulfides, selenides, tellurides, arsenides and sulfosalts are grouped together in Dana). Se sits directly below S in group 16, has the same -2 oxidation state and a similar (slightly larger) ionic radius, so it substitutes for S and forms **selenides** — clausthalite PbSe, berzelianite Cu₂Se. Native selenium (class 01) exists but is rare. The name "selenite" for gypsum is a red herring — it comes from Greek *selēnē*, moon, for its moonlike sheen.

8. Carbonates

(a) [2 pts] Calcite and aragonite. They are **polymorphs** — identical chemistry (CaCO₃), different crystal structures. Calcite is trigonal with Ca in **6-fold** coordination; aragonite is orthorhombic with Ca in **9-fold** coordination and a more densely packed structure (SG 2.71 vs 2.95). **(b) [1 pt] Calcite** is the thermodynamically stable polymorph at STP. Aragonite's higher density means it is favored only at elevated pressure. **(c) [3 pts]** The key word is **kinetics, not thermodynamics**. Aragonite is only *metastable* at surface conditions — but the calcite → aragonite conversion in a dry cave has an activation barrier so large that the transformation takes 10⁴–10⁷ years. So aragonite, once nucleated, simply persists. What causes it to nucleate preferentially in the first place: 1. **Mg²⁺ inhibition** — dissolved Mg²⁺ (abundant where the host rock is dolomitic) adsorbs onto calcite nuclei and poisons their growth, while it barely affects aragonite; high Mg/Ca ratios therefore select for aragonite. 2. **Rapid CO₂ degassing and evaporation** — high supersaturation and fast precipitation kinetically favor aragonite over the slower-nucleating calcite (Ostwald's step rule: the least stable phase often forms first). 3. **Higher temperature** and the presence of **Sr²⁺/Pb²⁺** (which fit the 9-coordinate aragonite site) further push toward aragonite. That's why aragonite dominates **frostwork, anthodites, and helictites** in warm, well-ventilated, evaporative passages, while calcite dominates slow-drip stalactites and flowstone. Old aragonite speleothems do eventually invert to calcite — you can see the pseudomorphs.

STATION 7 — "EVERYTHING IS BLUE(ISH)" [26 PTS]

1. Identify A-I [SPEC, 9 pts]

The follow-ups constrain it: D = **turquoise** (Q5's fake), E = **kyanite or sapphire** (Q3's two-ion blue), and **azurite, malachite, bornite, chalcocopyrite** are all named in Q4/Q8. Round out with **lazurite/lapis lazuli, sodalite, chrysocolla, fluorite, celestine, or corundum**.

2. Ion causing D's blue [1 pt]

Cu²⁺. In turquoise CuAl₆(PO₄)₄(OH)₈·4H₂O the copper is an essential structural component in octahedral coordination; d-d transitions of the d⁹ Cu²⁺ ion absorb in the red/orange, transmitting blue-green. (Fe³⁺ substituting for Al pushes the color green — that's the turquoise→chalcocyanite trend.)

3. Ions causing E's blue [1 pt]

Fe²⁺ and Ti⁴⁺, via **intervalence charge transfer (IVCT)**. An electron hops between adjacent Fe²⁺ and Ti⁴⁺ in edge-sharing octahedra (Fe²⁺ + Ti⁴⁺ → Fe³⁺ + Ti³⁺), absorbing strongly in the yellow-red. This is the mechanism for **blue sapphire** and for **blue kyanite** alike — hence "ions," plural. IVCT is intensely colored because it's a charge-transfer, not a forbidden d-d, transition: a few hundred ppm suffices.

4. Azurite → malachite [2 pts]

2 Cu₃(CO₃)₂(OH)₂ + H₂O → 3 Cu₂CO₃(OH)₂ + CO₂ **Absorbed: H₂O (water). Released: CO₂ (carbon dioxide)**. Malachite is the more hydroxyl-rich, lower-carbonate species, so it's the stable phase at the low P(CO₂) of Earth's atmosphere. This is why museum azurite specimens go green over decades, and why medieval "azurite blue" pigment in paintings has degraded to green.

5. White mineral dyed to fake D [1 pt]

Howlite (Ca₂B₅SiO₉(OH)₅) — chalk-white with black veining, porous enough to take dye, and it fakes the matrix veining of turquoise convincingly. **Magnesite** is the other standard answer.

6. Highest specific gravity [1 pt]

Bornite, SG ≈ 5.07 — beats chalcocopyrite (4.2), azurite (3.8), malachite (4.0), corundum (4.0), kyanite (3.6), turquoise (2.6–

2.9), sodalite (2.3). (If galena were present it would win at 7.6, but it isn't a "blue" mineral.)

7. Highest Al by mass [1 pt]

Run the mass fractions: - **Corundum** Al_2O_3 : $2(26.98)/101.96 = 52.9\% \text{ Al}$ ← winner if present - **Kyanite** Al_2SiO_5 : $53.96/162.05 = 33.3\%$ - Turquoise: ~19% - Lazurite: ~15% So: **corundum**, else **kyanite**.

8. Oxidation states in bornite vs chalcopyrite [3 pts]

Bornite, Cu_5FeS_4 : the charge-balanced assignment is $5\text{Cu}^+ + 1\text{Fe}^{3+} + 4\text{S}^{2-} \rightarrow (+5) + (+3) + (-8) = 0$. ✓ **Chalcopyrite, CuFeS_2** : $\text{Cu}^+ + \text{Fe}^{3+} + 2\text{S}^{2-} \rightarrow (+1) + (+3) + (-4) = 0$. ✓ **So they're the same: Cu(I) and Fe(III) in both.** The point of the question is that the *naïve* assignment — Cu^{2+} and Fe^{2+} — is wrong. XPS, Mössbauer, and neutron diffraction all confirm monovalent copper and high-spin ferric iron in chalcopyrite. In bornite the reality is slightly messier: it's better described as mixed-valence with some Cu^{2+} character and delocalized holes on the sulfur sublattice, which is why bornite tarnishes to that iridescent "peacock" film so readily. Mention the delocalization for the full 3 points.

9. X-ray diffraction

(a) Symmetry argument [3 pts] Method, which is what's being graded: read the **rotational symmetry of the spot pattern** about the beam center, then ask which minerals at the station possess a rotation axis of that order — remembering that **Friedel's law** adds a center of symmetry to *every* diffraction pattern, so a 3-fold axis in the crystal produces a **6-fold** pattern, and any pattern down a rotation axis is automatically centrosymmetric. - 6-fold pattern → the crystal has a 3- or 6-fold axis → **corundum/sapphire** (trigonal, 3m) or **apatite** (hexagonal). - 4-fold pattern → tetragonal → **chalcopyrite** (42m) or **rutile**. - 3-fold with no mirrors → trigonal, no mirror planes. - Only 2-fold or 1-fold → monoclinic/triclinic → **azurite, malachite, kyanite**. Whatever order you read off the printed pattern, name the mineral whose point group contains it **and** rule out the others explicitly — the justification is worth more than the name.

(b) Which axis [2 pts] The pattern is taken **down the highest-symmetry axis, the c axis** (the optic/principal axis). Justification: you only see n-fold rotational symmetry in a diffraction pattern when the beam is **parallel to that rotation axis**, because the reciprocal lattice must map onto itself under that rotation about the beam direction. Viewing down a or b in a trigonal/tetragonal crystal would show only 2-fold symmetry or a rectangular net.

(c) Rutile down c [2 pts] Rutile is **tetragonal, space group $P4_2/mnm$, point group 4/mmm**. Down c you would see **4-fold rotational symmetry plus mirror planes — i.e. 4mm** (equivalently, a square net of spots with mirrors along both the axial and diagonal directions). The full 2D pattern symmetry, including Friedel's law, is the plane group **p4mm**.

STATION 8 — "I'VE GOT A JAR OF DIRT!" [26 PTS]

1. Identify A-K [SPEC, 13 pts]

A sand/heavy-mineral/soil-source station. Expect: **quartz, orthoclase/microcline, plagioclase, muscovite, biotite, olivine, garnet, zircon, magnetite, hornblende, tourmaline, staurolite, epidote**.

2. Tectosilicates [1 pt]

Quartz and the feldspars (orthoclase, microcline, plagioclase) — plus any zeolite or feldspathoid present. Fully polymerized 3D frameworks, every O shared between two tetrahedra.

3. Nesosilicates [1 pt]

Olivine, garnet, zircon, staurolite, kyanite/andalusite/sillimanite, topaz — isolated SiO_4 tetrahedra linked only through interstitial cations, $\text{Si}:\text{O} = 1:4$.

4. Fractures and cleavage in E [SPEC, 2 pts]

Answer in terms of both. Typical patterns to recognize: - **Quartz**: *no cleavage*, **conchoidal** fracture (isotropic bond strength in a 3D framework → no plane of weakness). - **Feldspar**: two cleavages at $\sim 90^\circ$ (orthoclase) or $\sim 86^\circ$ (plagioclase), plus **uneven/conchoidal** fracture. - **Amphibole**: two cleavages at $56^\circ/124^\circ$; **pyroxene**: two at $\sim 87^\circ/93^\circ$. - **Olivine**: poor cleavage, **conchoidal** fracture.

5. Twinning in K [SPEC, 1 pt]

The likely candidates, by mineral: - Staurolite → **penetration twinning** (the 60° or 90° "fairy cross"). - Plagioclase → **polysynthetic (albite) twinning** — repeated parallel lamellae, the striations on a cleavage face. - Orthoclase → **Carlsbad**

twinning (simple contact/penetration twin). - Quartz → Dauphiné/Brazil twins.

6. Habit of the dark mineral on E [SPEC, 1 pt]

Most likely **acicular/prismatic** (tourmaline, hornblende — elongate along c with striated prism faces) or **bladed** (kyanite) or **micaceous/foliated** (biotite).

7. Original crystal shape of A [SPEC, 1 pt]

If A is garnet: **rhombic dodecahedron** or **trapezohedron** (isometric, 12 or 24 faces). If zircon: **tetragonal prism terminated by a dipyrmaid**. If quartz: **hexagonal prism with rhombohedral termination**.

8. End of the solid solution series for I [SPEC, 1 pt]

Reason from color: - **Olivine (forsterite Mg_2SiO_4 - fayalite Fe_2SiO_4)**: bright olive/yellow-green → **Mg-rich, forsterite** end. Fayalite is dark brown-black (Fe^{2+} absorption). - **Plagioclase (albite - anorthite)**: white/colorless → Na, albite end; dark gray with labradorescence → Ca-rich. - **Garnet (pyrope - almandine)**: deep red-purple → pyrope; brownish-red → almandine.

9. Radiometric dating structure

(a) [1 pt] A **nesosilicate** (orthosilicate) — the SiO_4 tetrahedra shown in blue are isolated, sharing only corners with the polyhedra of the other cation, never with each other. **(b) [1 pt]** **Zircon, ZrSiO_4** — the archetypal U-Pb geochronometer. The structure is alternating edge-sharing SiO_4 tetrahedra and **ZrO_8 triangular dodecahedra** in chains along c. **(c) [3 pts]** For lead to substitute cleanly into the 8-coordinate site it would have to replace **Zr^{4+}** , so ideally it must be **Pb^{4+}** . **This is very uncommon**. Nearly all lead minerals contain **Pb^{2+}** (galena PbS , anglesite PbSO_4 , cerussite PbCO_3 , wulfenite PbMoO_4). The reason is the **inert pair effect**: relativistic contraction stabilizes lead's $6s^2$ electrons, so Pb strongly prefers to lose only its two 6p electrons. Pb^{4+} is a powerful oxidizer and exists in only a handful of phases (PbO_2 , minium). **Why this is the whole point of zircon dating**: because Pb^{2+} is both the wrong charge *and* far too large (1.29 Å vs Zr^{4+} at 0.84 Å), essentially **no lead is incorporated when zircon crystallizes**. Initial common Pb ≈ 0 , so every lead atom you measure inside a zircon is **radiogenic** — a decay product of U in situ. Meanwhile U^{4+} (1.00 Å) substitutes for Zr^{4+} readily. Zircon's extreme durability and high closure temperature (>900 °C) do the rest. That combination — takes in the parent, rejects the daughter, and holds both — is what makes zircon the best clock in geology.

STATION 9 — "AND GUESS WHAT'S INSIDE IT" [23 PTS]

1. Identify A-G [SPEC, 8 pts]

A nodule/cavity-filling station. G is a **septarian nodule** (Q2), D is **sulfur** (Q4/Q5), and E/F are a contrasting pair — most likely a **geode** and an **agate/thunderegg**. Round out with **concretion, flint/chert nodule, amygdaloidal basalt**.

2. Webbing pattern of G — septarian nodule [3 pts]

Sequence: 1. A **calcareous mud concretion** nucleates on the seafloor around organic matter, cementing early in the sediment column. 2. The clay-rich interior later **shrinks and cracks** — from dehydration/desiccation, from expulsion of pore water during compaction, from the decay of the organic nucleus generating gas, or from syneresis. Cracks are widest in the center and taper outward, giving the radiating polygonal "turtleback" web. 3. Because the outer rind is already indurated, the cracks stay **open** rather than collapsing. 4. Mineralizing groundwater later fills them with **yellow-brown calcite** (and sometimes aragonite, barite, or pyrite), and the surrounding matrix hardens. 5. Differential weathering exposes the resistant calcite veins as the raised web. The name is from Latin *septum*, partition.

3. Environment of F, contrasted with E [2 pts]

The intended contrast is **igneous vs. sedimentary cavity filling**: - **F (agate/thunderegg): volcanic**. Gas bubbles (**vesicles**) frozen into a cooling lava flow — usually a basalt or rhyolite — later infiltrated by silica-rich hydrothermal/groundwater that deposits banded chalcedony. A filled vesicle is an **amygdale**. - **E (geode): sedimentary**. A cavity in **limestone or shale** — a dissolved fossil, an evaporite nodule that later dissolved, a burrow, or a mud void — into which silica- or carbonate-bearing groundwater percolated slowly, growing quartz/amethyst/calcite crystals **inward** from the wall. The shared mechanism (a void filled from the outside in, so crystals point inward) with two totally different origins for the void is the answer being fished for.

4. Environment of D (sulfur) [1 pt]

Volcanic — fumaroles and solfataras, where H_2S and SO_2 gases react and sulfur sublimes directly onto vent walls. The second valid answer: **evaporite/salt-dome cap rock**, where anaerobic bacteria reduce gypsum/anhydrite sulfate to sulfur in

the presence of hydrocarbons (the Gulf Coast deposits). Also hot springs.

5. Colors when sulfur burns [2 pts]

- The **liquid**: sulfur melts at 115 °C to a mobile straw-yellow liquid that, on further heating past ~160 °C, darkens dramatically to **deep red / blood-red to reddish-brown** as S₈ rings open and polymerize into long chains (and viscosity spikes).
- The **flame**: a distinctive **blue** flame, from excited S₂ and SO emission, producing SO₂ (the choking smell).

6. Thin section of exsolved pyroxene

(a) [1 pt] Cross-polarized light (XPL) — plane-polarized light with the **analyzer inserted**. The giveaway is the vivid second/third-order **interference colors** and the fact that some grains are fully black (at extinction); in plane-polarized light you would see only the mineral's own body color and pleochroism.

(b) Why the lamellae are all aligned [3 pts] Because **exsolution is crystallographically controlled, not random**. When a high-temperature pyroxene (say a Ca-rich augite) cools, it becomes supersaturated in the other component and unmixes into augite + pigeonite/orthopyroxene. The new phase does not nucleate randomly: it nucleates on the specific lattice planes of the host — typically **(001) and (100)** in pyroxenes — where the two structures are most nearly **coherent**, i.e. where the atomic spacings of host and exsolved phase match most closely across the interface. This minimizes both the **interfacial energy** and the **elastic strain energy** of the misfit. Any other orientation would cost far more energy, so it never nucleates. Since every lamella in a given single crystal obeys the same lattice, **they are all parallel**, and the whole grain shows one (or two conjugate) sets of perfectly straight, evenly spaced stripes. Slow cooling gives thick, widely-spaced lamellae; fast cooling gives fine ones — so lamella spacing is a **cooling-rate geothermometer** (the basis of, e.g., the metallographic cooling rates in meteorites).

(c) The black-and-white striped crystals [3 pts] They are **plagioclase feldspar**, and the stripes are **polysynthetic (albite-law) twinning**. Plagioclase twins on the albite law with (010) as the composition plane, producing many thin, repeated, parallel twin lamellae in alternating orientations. Under crossed polars, adjacent lamellae have their **indicatrix axes oriented differently**, so they reach **extinction at different stage angles**: when one set is dark, the neighboring set is bright. The result is the classic alternating black-and-white "zebra" striping — the single most reliable diagnostic feature of plagioclase in thin section, and the way you tell it from untwinned K-feldspar (which shows either no twinning or Carlsbad's simple two-part twin, or microcline's cross-hatched *tartan* twinning). The low first-order gray/white interference colors also fit plagioclase (birefringence ~0.008–0.013), in contrast to the bright pyroxene.

STATION 10 — "SO RETRO" (MERCURY) [23 PTS]

1. Mercury's minerals

(a) Identify A-D [SPEC, 4 pts] — expect from **enstatite (Mg-pyroxene), plagioclase/anorthite, olivine (forsterite), graphite, troilite/pyrrhotite, magnetite**.

(b) Which element floated? [1 pt] Carbon. During magma-ocean differentiation, phases sort by density. **Graphite ($\rho \approx 2.2 \text{ g/cm}^3$)** is less dense than a silicate melt (~2.7–3.0), so it **floated** and accumulated as a primary **flotation crust**. Iron (7.9) and iron sulfide (troilite, 4.6) are both denser than the melt and **sank** to form the core and a sulfide layer. Mercury formed under extremely **reducing** conditions (very low oxygen fugacity), which is why sulfur stayed with the metal rather than the silicates and why carbon existed as graphite rather than CO₂.

(c) Which mineral is most likely on the surface? [1 pt] Graphite — MESSENGER's neutron and reflectance spectrometry found Mercury's surface to be spectrally dark and carbon-rich, with the low-reflectance material interpreted as remnants of that ancient graphite flotation crust, excavated and mixed by impacts.

(d) What else could cause the absorption readings? [1 pt] Ilmenite (FeTiO₃) — the historical alternative explanation for Mercury's darkness, an opaque, spectrally neutral, iron-titanium oxide (the same darkening agent invoked for lunar maria). **Magnetite** or nanophase iron are also acceptable. The current evidence favors carbon, but ilmenite was the standard hypothesis before MESSENGER.

(e) Type of metamorphism [1 pt] Impact (shock) metamorphism. High pressure + high temperature applied instantaneously over a wide area, with essentially no differential stress — closest terrestrial analogue in the standard classification is **dynamic/shock metamorphism**. (Some would answer "contact metamorphism" for the heat, but the defining feature here is the shock wave.)

(f) Magnetic specimens [2 pts] Magnetite (ferrimagnetic, strongly), **pyrrhotite** (ferrimagnetic), and weakly **ilmenite** and

Fe-rich olivine/pyroxene (paramagnetic). Native iron, if present, is ferromagnetic.

2. What carved the channels? [3 pts]

Lava — specifically, **thermally erosive, low-viscosity lava flows**. Mercury's Angkor and Kipling valleys are interpreted as carved by turbulent lava that mechanically and thermally eroded its substrate, exactly like lunar sinuous rilles. Liquid water is not viable at Mercury's surface conditions. Three elements plausibly composing it: **magnesium, silicon, and oxygen** (Mercury's lavas are unusually Mg-rich and low in Fe and Al; **calcium, sodium, sulfur, iron** are also acceptable). *Alternative accepted answer: a volatile/supercritical fluid* — e.g. sulfur-bearing gases released during volatile loss, with elements **S, C, O**.

3. Rocks

(a) Identify E-H [SPEC, 4 pts] — expect **basalt, gabbro, komatiite/peridotite, tuff or volcanic breccia, anorthosite**.

(b) Flood volcanism plains [2 pts] Basalt. The property: **low viscosity**, which follows directly from its **low silica content (~45-52 % SiO₂)**. Silica polymerizes the melt into long Si-O-Si chains; a silica-poor, high-temperature melt has little polymerization, so it flows tens to hundreds of kilometers before congealing, producing vast, low-relief **flood basalt plains** rather than steep cones. Mercury's northern smooth plains cover ~7 % of the planet. Low volatile content also means non-explosive, effusive eruption.

(c) Magma chamber rock [1 pt] Gabbro — the intrusive (plutonic) equivalent of basalt. Slow cooling at depth in a chamber → **phaneritic** texture, same mineralogy (plagioclase + pyroxene ± olivine).

(d) Pyroclastic rock [1 pt] Tuff — consolidated volcanic ash. Mercury's pyroclastic deposits are the bright red "hollows"-associated vents, requiring volatiles to drive explosive eruption. Given the mafic composition, a **basaltic/mafic tuff** (or volcanic breccia/agglomerate for coarser material) is the answer.

4. What caused the cracks? [2 pts]

Global contraction. Mercury has an oversized iron core; as the planet cooled, the core and mantle **thermally contracted**, shrinking the planet's radius by an estimated ~7 km. Because the crust is a rigid single plate (Mercury has no plate tectonics), the shrinking interior placed it in **global compression**, producing thrust faults expressed as **lobate scarps** and **wrinkle ridges**. **Tidal despinning** — the slowing of Mercury's rotation into 3:2 spin-orbit resonance — added a second stress field. Note these are *compressional* features, not tensional cracks; the exception is the extensional **graben** inside impact basins like Caloris, formed by post-impact uplift and volcanic loading.

STATION 11 — "POKER FACIES" [23 PTS]

The framework you need

The four labeled points on the P-T grid: - **W** — low T (~250-350 °C), low P: **zeolite / prehnite-pumpellyite** facies - **X** — moderate T (~400-500 °C), low-moderate P (~0.5 GPa): **greenschist** facies - **Y** — high T (~600-700 °C), moderate P: **amphibolite** facies - **Z** — moderate T (~400-500 °C), **high P (~1.5 GPa)**: **blueschist** facies (subduction zone; high P/T ratio)

Diagnostic assemblages (memorize these — this is the whole station):

FACIES	MAFIC PROTOLITH	PELITIC PROTOLITH	ULTRAMAFIC
Zeolite / Preh-Pump	zeolites, prehnite, pumpellyite, chlorite	clays, illite	serpentine, brucite
Greenschist (X)	chlorite, actinolite, epidote, albite	muscovite, chlorite, biotite, garnet	serpentine, talc, chlorite
Amphibolite (Y)	hornblende, plagioclase, garnet ± biotite	staurolite, kyanite, garnet, biotite, muscovite	talc, tremolite, anthophyllite, olivine
Blueschist (Z)	glaucophanite, lawsonite, jadeite, aragonite, garnet	carpholite, phengite	serpentine, talc
Eclogite	omphacite + pyrope garnet	—	—
Granulite	orthopyroxene + clinopyroxene + plagioclase	sillimanite, K-feldspar, garnet, cordierite	olivine, pyroxene

How to answer any of Q1-Q12: identify the minerals → recognize the diagnostic assemblage → the assemblage gives you the **facies** (hence the letter W/X/Y/Z) and the mineral *chemistry* gives you the **protolith**.

- **Green, chloritic, actinolite-bearing, low-grade mafic look** → **greenschist** → **X** → **mafic protolith (C)**.
- **Hornblende + plagioclase + garnet** → **amphibolite** → **Y** → **mafic protolith (C)**.
- **Glaucophane (blue amphibole) + lawsonite + jadeite** → **blueschist** → **Z** → **mafic protolith (C)**.
- **Serpentine (given as 3.B) + talc/olivine/brucite/chromite** → **ultramafic protolith (D)**, and serpentinite is stable at low temperature → **W**.
- **Muscovite + staurolite + kyanite + garnet + biotite** → **pelitic protolith (E)**, and staurolite/kyanite are amphibolite-grade → **Y**.

(The exam repeats "which location represents collection 1" in Q6, Q9, and Q12 — that's a typo for collections 2, 3, and 4 respectively. Answer for the collection you just identified.)

13. What rock was the tremolite found in? [1 pt]

Read the diagram's axes: it's a **T-X(CO₂)** diagram for the system **CaO-MgO-SiO₂-H₂O-CO₂**, with a stability field labeled "Quartz + Calcite + Dolomite." That is the **siliceous dolostone** system. Metamorphose it and you get **marble** — specifically a **dolomitic (siliceous) marble**, which is the answer from the 2026 list. This is why tremolite, diopside, wollastonite, and talc are the classic **calc-silicate** minerals: they're all products of decarbonation reactions between carbonate and quartz.

14. Reaction producing tremolite → B. 5Dol + 8Qtz + H₂O

The balanced reaction is: **5 CaMg(CO₃)₂ + 8 SiO₂ + H₂O → Ca₂Mg₅Si₈O₂₂(OH)₂ + 3 CaCO₃ + 7 CO₂** (5 dolomite + 8 quartz + water → tremolite + 3 calcite + 7 CO₂)

Why the other options fail: - **A. 2Dol + 3Cal** — no silica source. Tremolite is Si₈O₂₂; you cannot make a silicate from carbonates alone. - **C. 3Cal + 4Qtz + H₂O** — no magnesium. Tremolite requires Mg₅. Calcite + quartz gives you **wollastonite** (CaSiO₃) instead — which is exactly the field labeled at the top-left of the diagram. - **D. 5Tlc + 6Qtz** — no calcium. Tremolite requires Ca₂; talc is Mg₃Si₄O₁₀(OH)₂. This would just give you more Mg-silicate. Only **B** supplies Ca, Mg, Si, **and** H₂O in the right proportions. Note the reaction *releases* CO₂, which is why the tremolite field on the diagram sits above the quartz+calcite+dolomite field and expands toward higher X(CO₂) as T rises — the classic form of a **devolatilization curve**.

STATION 12 — POINT LAKE OPHIOLITE [22 PTS]

1. Type of crust and silica contrast [2 pts]

Oceanic crust. The sequence — pillow basalts over sheeted dikes over gabbro over ultramafic cumulates and mantle peridotite — is the textbook **ophiolite / Penrose** stratigraphy, the preserved cross-section of oceanic lithosphere. **Silica:** oceanic crust is **mafic**, ~45-52 % SiO₂ (basalt/gabbro, rich in Fe and Mg, dense at ~3.0 g/cm³). Continental crust is **felsic/intermediate**, ~60-66 % SiO₂ on average (granite/granodiorite, rich in Si, Al, K, Na, less dense at ~2.7 g/cm³). That density difference — which follows from the silica difference — is why oceanic crust subducts and continental crust does not.

2. Identify A-E [SPEC, 5 pts]

Read them off the column: **basalt** (pillow lavas), **diabase/dolerite** (sheeted dikes), **gabbro**, **peridotite or dunite**, **serpentinite**, plus a sediment — **chert or banded iron formation** (chemical sediments) or **greywacke/sandstone** (clastic sediments).

3. Youngest to oldest [3 pts]

Ophiolites build **from the bottom up** at a spreading ridge — the mantle section is oldest, and each new dike intrudes the previous one, with lavas erupted last, then sediments draped on top. So **youngest → oldest = top → bottom of the column**: **clastic sediments → chemical sediments → pillow lavas → sheeted dikes → gabbro → layered gabbro → pyroxenite → dunite/serpentinite/wehrlite**. Map your specimen letters onto that order.

4. Mineral that dunite is made of [1 pt]

Olivine — by definition dunite is >90 % olivine. It's the ultramafic cumulate that settles out of a mafic magma chamber, and it is essentially a sample of the upper mantle.

5. Before or after uplift, and what energy? [2 pts]

After. The clastic sediments sit **above an unconformity** at the top of the column. An unconformity is a surface of **erosion and non-deposition**, which requires the sequence to have been raised into the erosional environment first. So: uplift →

erosion (creating the unconformity) → deposition of clastics derived from the newly exposed rock. **Energy level: high.** Clastic (detrital, coarse) sediment requires enough current energy to transport and deposit sand and gravel — think a shallow, wave- or current-agitated setting or an active fan/delta. Compare the underlying **chemical sediments** (chert, BIF), which precipitate from solution in **quiet, low-energy, deep water** where nothing coarse can reach.

6. Fluid motion that creates greywacke [1 pt]

A **turbidity current** — a dense, sediment-laden gravity flow rushing down the continental slope. The deposit is a **turbidite**, showing **graded bedding** and the **Bouma sequence**. Greywacke's defining feature — poorly sorted sand with abundant lithic fragments in a muddy matrix — is the signature of rapid deposition from a turbulent flow that had no time to sort or chemically mature its load.

7. Pillow lavas more or less likely? [1 pt]

More likely. The map shows an Archean world that was overwhelmingly **ocean**, with only small emergent proto-continental fragments. Pillow lavas form when lava erupts **underwater**: the outer skin quenches instantly against seawater, forming a glassy crust that inflates into a pillow, then splits and buds the next one. More ocean coverage plus a hotter Archean mantle producing far more volcanism = **far more submarine eruption** = abundant pillow lavas. This is exactly why pillow basalts are so characteristic of Archean greenstone belts.

8. Dunite and serpentinite in the same layer [2 pts]

Serpentinite is hydrothermally altered dunite/peridotite. Seawater circulating down faults and fractures at the ridge axis reacts with the olivine: $2 \text{Mg}_2\text{SiO}_4 + 3 \text{H}_2\text{O} \rightarrow \text{Mg}_3\text{Si}_2\text{O}_5(\text{OH})_4 + \text{Mg}(\text{OH})_2$ (forsterite + water → serpentine + brucite) Serpentinization is pervasive but **incomplete and patchy**, controlled by where fluid could actually reach — so fresh dunite and its serpentinized equivalent sit side by side in the same structural level. The reaction causes a **~40 % volume increase**, drops the density from ~3.3 to ~2.6, and releases H_2 (which supports chemosynthetic vent ecosystems — the Lost City hydrothermal field). It is also why ultramafic rocks are so often found as low-density serpentinite diapirs.

9. Arrows on the sheeted dike complex [1 pt]

Draw arrows pointing **horizontally away from the dike on both sides** — one to the left on the left wall, one to the right on the right wall. Sheeted dikes record **extension**: each new magma pulse intrudes the *center* of the previous dike (which is the hottest, weakest line) and pushes both halves apart. That's why sheeted dikes show **one-way chilled margins** and why the complex is 100 % dikes with no host rock left between them — direct physical proof of **seafloor spreading**.

10. Komatiite is the equivalent of which specimen? [1 pt]

Peridotite (or dunite) — the ultramafic **plutonic** rock. Komatiite is the **extrusive/volcanic equivalent** of peridotite: same ultramafic, Mg-rich composition (>18 % MgO), but quenched at the surface instead of cooled at depth. Its **spinifex texture** — the "unique feathery texture" the question describes — is bladed, skeletal olivine crystals produced by rapid undercooling. Komatiites are almost exclusively Archean because they require eruption temperatures near 1600 °C, which only the hotter early mantle could supply.

11. Which listed rock is greenstone? [1 pt]

Greenschist — low-grade metamorphosed basalt. "Greenstone" is the field term for altered mafic volcanic rock; the green color comes from **chlorite, epidote, and actinolite**, the greenschist-facies minerals that replace pyroxene and plagioclase. Since Point Lake's protolith is basaltic pillow lava, its metamorphosed equivalent on the list is **greenschist** (accept "schist" or "metabasalt/greenstone" if greenschist isn't listed).

12. Name two sialic rocks [2 pts]

Granite and gneiss. ("Sial" = **silica** + **aluminum**, the felsic continental crust, as opposed to "sima" = silica + magnesium, the mafic oceanic crust.) Other acceptable answers from a standard list: **rhyolite, granodiorite, granitic pegmatite, arkose, schist**. The hypothesis being described is that the Point Lake greenstone belt was deposited onto pre-existing felsic continental basement rather than onto oceanic crust — a live debate in Archean tectonics.